

Perishable Lot–Sizing with Heterogeneous Shelf Lives

Compact MIP with Last–Expiry–First Allocation

Indices and sets

- $T = \{1, 2, \dots, \mathbf{T}\}$ — planning periods
- $I = \{1, 2, \dots, \mathbf{I}\}$ — set of items
- $t \in T$ — production period, $u \in T$ — consumption period
- m_{it} — shelf life of item i produced in period t (may vary by t)

For each item i and production period t , define:

$$\Gamma_{it} = \{u \in T : t \leq u \leq \min(T - 1, t + m_{it} - 1)\}$$

as the set of feasible consumption periods. For each (i, t) , define the expiry date:

$$v_{it} = t + m_{it}$$

Let Triples = $\{(i, t, u) : i \in I, t \in T, u \in \Gamma_{it}\}$.

Perishability

A lot produced at period t may be used only in periods $u \geq t$ satisfying $u - t \leq m_{it}$. Such a lot expires at period $t + m_{it}$.

Parameters

d_{iu}	demand of item i in period u
c_{it}	unit production cost of item i in period t
h_{it}	holding cost per unit of age (constant or time-varying)
s_{it}	setup cost for item i in period t
κ_t	total production capacity available in period t
p_{it}	optional per-item production capacity in period t
μ_{it}	big- M constant for setup linking, defined as $\mu_{it} := \sum_{u \in \Gamma_{it}} d_{iu}$
g_{itu}	cost of producing in t and consuming in u , namely $c_{it} + \sum_{\tau=t}^{u-1} h_{i\tau}$
C_{iu}	upper bound on the quantity allocated to demand (i, u) (typically $C_{iu} = d_{iu}$)
π_{iu}	penalty cost for lost sales of item i in period u (if allowed)

Decision variables

$X_{itu} \geq 0$	quantity of item i produced in period t and consumed in period u (defined for $(i, t, u) \in \text{Triples}$)
$Y_{it} \in \{0, 1\}$	binary setup decision for item i in period t
$Z_{itu} \in \{0, 1\}$	binary activation variable for arc (i, t, u)
$S_{itu} \in \{0, 1\}$	permission variable: $S_{itu} = 1$ if lot (i, t) is consumed at or after period u

Objective

$$\min \left[\sum_{(i,t,u) \in \text{Triples}} g_{itu} X_{itu} + \sum_{i \in I} \sum_{t \in T} s_{it} Y_{it} \right]$$

Constraints

Global capacity For all $t \in T$,

$$\sum_{i \in I} \sum_{u \in \Gamma_{it}} X_{itu} \leq \kappa_t$$

Setup linking For all $i \in I$ and $t \in T$,

$$\sum_{u \in \Gamma_{it}} X_{itu} \leq \mu_{it} Y_{it}$$

Demand satisfaction For all $i \in I$ and $u \in T$,

$$\sum_{t: u \in \Gamma_{it}} X_{itu} = d_{iu}$$

Arc activation For all $(i, t, u) \in \text{Triples}$,

$$X_{itu} \leq C_{iu} Z_{itu}$$

Permission variable definition For all $i \in I$ and $t \in T$ with $\Gamma_{it} \neq \emptyset$, let $u_{\max} = \max \Gamma_{it}$:

$$S_{itu_{\max}} = Z_{itu_{\max}} \tag{1a}$$

$$S_{itu} \geq S_{it(u+1)} \quad \forall u \in \{t, \dots, u_{\max} - 1\} \tag{1b}$$

$$S_{itu} \geq Z_{itu} \quad \forall u \in \Gamma_{it} \tag{1c}$$

$$S_{itu} \leq S_{it(u+1)} + Z_{itu} \quad \forall u \in \{t, \dots, u_{\max} - 1\} \tag{1d}$$

Last-Expiry-First-Out (LEFO) allocation For all $i \in I$, $t_1, t_2 \in T$ with $v_{it_1} < v_{it_2}$, and $u \in \Gamma_{it_1}$ with $u \geq t_2$ and $u < \min(v_{it_1}, \max \Gamma_{it_2})$:

$$Z_{it_1 u} + S_{it_2(u+1)} \leq 1$$

Variable domains

$$X_{itu} \geq 0, \quad Y_{it} \in \{0, 1\}, \quad Z_{itu} \in \{0, 1\}, \quad S_{itu} \in \{0, 1\}$$

Notes

- The permission variable S_{itu} indicates whether production batch (i, t) is consumed at or after period u
- The LEFO constraint ensures that earlier-expiring batches cannot be consumed after later-expiring batches
- The warehouse capacity constraint limits the total inventory carried from period u to $u + 1$
- The model uses a tight big- M formulation with $\mu_{it} = \sum_{u \in \Gamma_{it}} d_{iu}$ for setup linking